CMD-YOLO: An IoT-Oriented Lightweight Vision Framework for Real-Time Chili Maturity Detection and Robotic Localization on Edge Devices

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***Abstract*—Accurate crop maturity assessment and spatial localization are essential for intelligent robotic harvesting in Internet of Things (IoT)-based smart agriculture. However, real-world deployment is constrained by limited computational resources, complex illumination variations, dense target occlusion, and overlapping objects in unstructured agricultural environments. Considering the superior tradeoff between detection accuracy and inference efficiency of one-stage detectors, the YOLO architecture was selected as the baseline framework for edge-deployable agricultural robotic applications. To address these challenges, this article proposes CMD-YOLO, a lightweight edge-oriented framework for chili maturity detection and robotic localization. First, a high-fidelity chili maturity dataset containing multiple maturity stages under varying illumination, dense occlusion, and overlapping conditions was constructed to provide realistic visual characteristics for intelligent harvesting research. Built upon YOLOv11n, the proposed framework introduces a context-aware feature aggregation mechanism that combines local convolution with global attention to enhance feature representation capability for small and partially occluded chili targets. In addition, a detail-preserving downsampling strategy was designed to alleviate fine-grained spatial information loss during feature compression, thereby improving small-target perception and structural feature retention while maintaining lightweight computational complexity. Experimental results demonstrate that CMD-YOLO achieves 92.9% precision, 86.1% recall, 94.4% mAP50, and 75.3% mAP50–95 with only 3.66 M parameters. Deployed on an NVIDIA Jetson AGX Orin platform, the proposed model runs at 58.5 f/s in real time. Furthermore, robotic localization experiments integrating RGB-D perception and geometric coordinate transformation validate the practical applicability of the proposed framework, achieving a depth estimation error of only 4.4 mm for end-effector positioning. These results demonstrate that CMD-YOLO provides an efficient, robust, and edge-deployable solution for IoT-enabled intelligent robotic harvesting in complex agricultural environments.**

***Index Terms*—Agricultural robotics, edge intelligence, fruit maturity detection, Internet of Things (IoT), lightweight neural networks, object detection, smart agriculture.**

I. Introduction

With the rapid development of the Internet of Things (IoT) and edge intelligence, modern agricultural systems are evolving toward distributed, autonomous, and data-driven operation paradigms [1], [2], [3], [4], [5], [6], [7], [8], [9]. In IoT-enabled smart agriculture, large-scale deployment of resource-constrained sensing and perception nodes requires real-time, reliable, and energy-efficient visual intelligence to support automated decision-making and robotic actuation [10], [11], [12], [13]. As one of the world's leading producers of chilies, China is currently witnessing a transformation of its chili industry toward intelligent and precision agriculture, where high-quality harvesting and fine-grained maturity control are becoming increasingly critical [14], [15], [16]. However, traditional harvesting methods remain heavily dependent on manual labor, exhibiting low efficiency and high costs. Existing mechanical harvesting solutions, which typically rely on nonselective stripping mechanisms, often result in high fruit damage rates, inability to distinguish maturity levels, and significant impurity introduction, thereby severely compromising product quality [17], [18], [19]. Consequently, developing IoT-oriented intelligent harvesting systems with accurate visual perception capabilities has become a crucial pathway for agricultural modernization [20].

Within such IoT-driven agricultural systems, precise fruit maturity recognition serves as a fundamental prerequisite for enabling selective harvesting and intelligent decision-making [21], [22], [23], [24]. However, real-world agricultural environments are inherently unstructured and present substantial challenges for reliable perception. Chili plants typically exhibit asynchronous growth patterns, where green, semi-red, and fully red fruits coexist on a single plant. In addition, the detection process is significantly affected by variations in fruit size, color similarity between fruits and foliage, dense distribution, severe occlusion, and dynamic environmental

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conditions such as illumination changes and leaf interference [25]. These factors jointly lead to degraded detection accuracy and robustness, particularly under edge deployment scenarios with limited computational resources.

Early attempts to address fruit maturity detection primarily relied on traditional image processing and handcrafted feature extraction methods, such as color histograms, texture descriptors, and shape-based features. For instance, Karki et al. [26] combined multiple color spaces (RGB, HLS, CIELab, and YCbCr) with biological features and applied Gaussian naïve Bayes (GNB), support vector machines (SVMs), and feedforward artificial neural networks (FANNs) for strawberry maturity classification, where FANNs achieved the best performance in the CIELab space. Nevertheless, such approaches are highly sensitive to illumination variations and background interference, and they heavily depend on manually designed features, resulting in limited robustness and poor generalization capability in complex field environments [27], [28].

To overcome the limitations of handcrafted feature-based methods, recent studies have increasingly focused on deep-learning-based object detection and instance segmentation frameworks for crop maturity recognition under complex agricultural environments. Li et al. [29] proposed an improved mask region-based convolutional neural network (Mask R-CNN) model for grape maturity detection and segmentation of "Kyoho" grape bunches under complex orchard environments. Their method enhanced maturity-level recognition accuracy through refined instance segmentation; however, the model still suffered from relatively high computational complexity and limited real-time performance. Wang et al. [30] developed a Transformer-based Mask R-CNN framework for tomato detection and segmentation, which improved global contextual feature extraction and occlusion robustness in dense agricultural scenes. Nevertheless, the Transformer architecture introduced a large number of parameters and increased computational cost, making deployment on embedded robotic platforms challenging. Parvathi and Selvi [31] utilized Faster R-CNN for coconut maturity-stage detection in complex natural backgrounds and achieved favorable recognition accuracy, but the two-stage detection pipeline resulted in slower inference speed and reduced real-time applicability. Liu et al. [32] proposed the SE-Mask R-CNN model for apple detection and segmentation by integrating squeeze-and-excitation attention mechanisms to improve feature representation. Although the method enhanced detection performance in orchard environments, the instance segmentation framework still imposed considerable computational overhead. Similarly, Shiu et al. [33] applied Faster R-CNN and Mask R-CNN to pineapple detection and segmentation in UAV imagery, demonstrating good detection capability under aerial-view agricultural scenarios, while inference efficiency remained limited for real-time robotic harvesting tasks. Qi et al. [34] introduced an improved Mask R-CNN approach for papaya maturity detection using multitarget sampling strategies, which improved recognition performance in overlapping fruit scenarios but still relied on computationally intensive segmentation operations. Compared with these non-YOLO approaches, YOLO-based one-stage detectors provide a superior balance between detection accuracy, inference speed, and lightweight deployment, making them more suitable for real-time crop maturity recognition and robotic harvesting applications in resource-constrained agricultural environments.

Driven by the demand for real-time perception and edge deployment in intelligent agricultural systems, YOLO-based one-stage detectors have recently attracted increasing attention due to their favorable tradeoff between detection accuracy, inference speed, and computational efficiency. Consequently, a growing number of studies have explored improved YOLO frameworks for crop detection and maturity recognition tasks under complex agricultural environments [35], [36], [37], [38].

Beyond chili detection, maturity recognition has also been extensively studied in other crops. Li et al. [29] combined Mask R-CNN with attention mechanisms for grape cluster segmentation and maturity assessment, achieving an average accuracy of 94.4%. Cong et al. [39] improved Mask R-CNN for multiclass sweet pepper detection, achieving AP and AR of 98.1% and 99.4%, respectively. Huang et al. [40] optimized YOLOX-S for apple maturity recognition, improving mAP by 1.97%. Chen et al. [41] developed MTD-YOLOv7 for joint detection of cherry tomato fruits, bunches, and maturity, achieving an overall accuracy of 80% with an inference time of 4.9 ms. While Xu et al. [42] proposed the lightweight YO-RFEW model, achieving 90.82% mAP for multiclass melon maturity detection with a 1.5-ms inference speed. Zhu et al. [43] proposed Olive-EfficientDet, achieving mAP between 93.75% and 96.05% for multiple olive varieties. Furthermore, lightweight model design and mobile deployment have become increasingly important. Chen et al. [44] developed CES-YOLOv8 for strawberry maturity detection (92.1% accuracy). Si et al. [45] proposed Chili-YOLO, achieving 88.9% mAP in open-field chili maturity classification, although cross-dataset generalization was not validated.

In addition to maturity recognition, robotic localization has become a critical component in intelligent agricultural harvesting systems, as accurate spatial positioning directly determines the success rate of robotic grasping and picking operations. Recent studies have explored RGB-D perception, depth estimation, and coordinate transformation methods for agricultural robotic localization tasks. For example, Sa et al. [46] combined RGB-D sensing with deep visual perception for autonomous sweet pepper harvesting, enabling accurate fruit localization under occlusion conditions. Zheng et al. [47] proposed a binocular vision-based localization framework for tomato harvesting robots, improving depth estimation accuracy in complex greenhouse environments. Liu et al. [48] integrated visual detection with robotic coordinate transformation to achieve real-time fruit position estimation for robotic grasping tasks. In addition, several studies introduced eye-in-hand and eye-to-hand calibration strategies to improve the spatial mapping accuracy between camera and robotic manipulators [49], [50]. Although these methods achieved promising localization performance, many existing studies mainly focused on fruit positioning accuracy while paying less attention to the collaborative optimization between maturity recognition,

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lightweight perception models, and edge deployment efficiency under unstructured agricultural environments. Therefore, developing an integrated lightweight framework that simultaneously considers maturity detection and robotic localization remains a significant challenge for intelligent agricultural harvesting systems.

Despite these advancements, several critical challenges remain unresolved. First, most existing studies focus on coarse-grained classification (e.g., binary maturity), while lacking effective characterization of multistage maturity variations under real agricultural conditions. Second, many approaches prioritize accuracy improvement without explicitly considering edge deployment constraints, including computational complexity, memory consumption, inference latency, and power efficiency, which restricts their applicability in IoT-enabled agricultural systems. In addition, existing robotic localization methods mainly emphasize spatial positioning accuracy, whereas the coordination between maturity recognition, target localization, and robotic harvesting execution remains insufficiently investigated. Third, current models often exhibit insufficient robustness under complex field conditions, particularly in scenarios involving dense occlusion, illumination variation, target overlap, and dense object distribution. Finally, balancing lightweight architecture design, detection accuracy, and real-time localization capability still remains challenging for edge-based intelligent harvesting applications.

Despite these advancements, several critical challenges remain unresolved. First, most existing studies focus on coarse-grained classification (e.g., binary maturity), while lacking effective characterization of multistage maturity variations under real agricultural conditions. Second, many approaches prioritize accuracy improvement without explicitly considering edge deployment constraints, including computational complexity, memory consumption, inference latency, and power efficiency, which restricts their applicability in IoT-enabled agricultural systems. In addition, existing robotic localization methods mainly emphasize spatial positioning accuracy, whereas the coordination between maturity recognition, target localization, and robotic harvesting execution remains insufficiently investigated. Finally, current models often exhibit insufficient robustness under complex field conditions, particularly in scenarios involving dense occlusion, illumination variation, target overlap, and dense object distribution. Finally, balancing lightweight architecture design, detection accuracy, and real-time localization capability still remains challenging for edge-based intelligent harvesting applications.

To tackle the four above-mentioned bottlenecks simultaneously, this article designs three targeted network optimization components and integrates them into the lightweight YOLOv11n baseline: first, an EfficientViT hybrid backbone is introduced to fuse local convolution and global linear attention, solving the problem of insufficient long-range context modeling for occluded small chili fruits; second, the C3k2_ConvFormer neck module is constructed to combine self-calibrated convolution and Transformer feature interaction, enhancing the feature differentiation ability of ambiguous transitional maturity fruits; third, a progressive spatial convolution (PSConv) detail-preserving downsampling convolution module is proposed to retain subtle color gradient features of ripening chilies and mitigate feature attenuation in the compression process. Combining the three improved modules, we propose CMD-YOLO, a lightweight maturity detection framework oriented to IoT agricultural edge terminals.

The proposed framework is designed from an IoT-oriented perspective, aiming to achieve a balanced optimization of detection accuracy, computational efficiency, and deployment adaptability for edge-enabled agricultural systems. The main contributions of this work are summarized as follows.

1) A high-fidelity chili maturity dataset was established under complex field conditions, covering multiple maturity stages with varying illumination, dense occlusion, and overlapping targets. The dataset provides realistic visual characteristics for robust chili maturity detection and robotic harvesting research.
2) An IoT-oriented lightweight detection framework, termed CMD-YOLO, was developed for real-time chili maturity detection and robotic localization on edge devices. Unlike conventional lightweight modifications of YOLO-based detectors, the proposed framework jointly considers detection accuracy, computational efficiency, and deployment adaptability for agricultural robotic systems.
3) A context-aware feature aggregation mechanism integrating local convolution and global attention was introduced to enhance the representation capability of small, dense, and partially occluded chili targets in unstructured agricultural environments.
4) A detail-preserving downsampling strategy was designed to reduce fine-grained feature information loss during spatial compression, thereby improving feature retention and small-target perception performance.
5) The proposed framework was deployed and validated on an embedded edge-computing robotic platform, demonstrating favorable robustness and practical applicability for real-world intelligent agricultural harvesting scenarios.

The remainder of this article is organized as follows. Section II introduces image collection procedures and the construction scheme of the chili dataset. Section III elaborates the overall architecture of the proposed CMD-YOLO model as well as the detailed design of the EfficientViT backbone, C3k2_ConvFormer, and PSConv core modules. Section IV conducts comprehensive multidimensional analysis including comparative experiments, ablation studies, heatmap visualization, edge deployment tests, and robotic localization measurement. Section V summarizes all research outcomes and points out directions for subsequent optimization research.

II. Experimental Materials

A. Image Acquisition

Dongshan Guangpi chili pepper, a representative local cultivar widely cultivated in Huangxing Town and Lituo Township of Changsha County, Hunan Province, China, was selected as the research object. This cultivar is characterized by thick

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Fig. 1. Examples of field image acquisition and maturity-stage classification for chili pepper fruits.

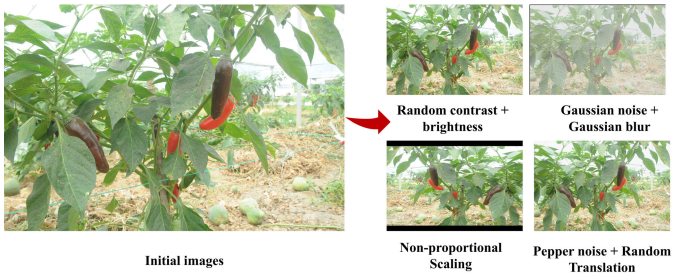


Fig. 2. Examples of augmented chili pepper fruit images.

flesh, smooth epidermis, mild pungency, and favorable storage properties, making it suitable for both fresh-market consumption and mechanized harvesting applications.

Image acquisition was conducted from July to September 2023 at the chili pepper cultivation base of Duolai Agricultural Technology Company, Ltd., Changsha County, China. A Canon M100 camera was used to collect field images under natural growth conditions. According to fruit color and phenological characteristics, chili pepper fruits were categorized into three maturity stages: future mature (FM), corresponding to the green immature stage; partially mature (PM), corresponding to the color-transition stage from brown to red; and ideal mature (IM), corresponding to the fully red harvest stage.

Considering the variability of field environments, image acquisition was performed under different illumination conditions, including sunny and cloudy weather, as well as different time periods throughout the day. Meanwhile, images were captured from multiple viewing angles and distances to simulate the visual perception characteristics of robotic harvesting systems. The collection scenarios and maturity classification criteria are illustrated in Fig. 1.

Unlike conventional image collection approaches that emphasize ideal viewpoints, the constructed dataset intentionally retained samples with dense occlusion, overlapping targets, backlighting, and pose variation. This strategy enables the dataset to better reflect the visual characteristics encountered in practical robotic harvesting environments. After manual screening, blurred images and samples with severe target loss were removed, resulting in an original dataset containing 1069 high-quality images.

B. Dataset Construction and Augmentation

To increase data diversity and reduce the risk of model overfitting, a primary data augmentation process was applied to the original dataset, including Gaussian noise addition, image blurring, rotation transformation, brightness adjustment, and random cropping. After augmentation, the dataset expanded to 3189 images.

In practical robotic harvesting scenarios, the visual system is further affected by motion-induced viewpoint variation, local overexposure, weak illumination, and complex foreground occlusion. Therefore, a secondary augmentation strategy was further introduced to improve the adaptability of the detection model to challenging field conditions. A total of 300 images were randomly selected from the augmented dataset and processed using combined augmentation operations, including random contrast and brightness adjustment, Gaussian noise with Gaussian blur, nonuniform scaling, and salt-and-pepper noise with random translation. This process generated an additional 1200 images.

Consequently, the final dataset contained 4389 annotated images of chili pepper fruits across different maturity stages. The X-AnyLabeling tool was employed for fine-grained annotation of the FM, PM, and IM categories. Finally, the dataset was divided into training, validation, and test sets at a ratio of 8:1:1, corresponding to 3511, 439, and 439 images, respectively. Examples of the augmented images are shown in Fig. 2.

III. Proposed Methodology

A. Technical Roadmap and Research Framework for the Chili Maturity Classification System

As illustrated in Fig. 3, this study proposes an end-to-end technical framework for chili fruit maturity classification, specifically designed to address the challenges of integrated recognition and localization in unstructured field environments. Specifically, the visual perception task in this study is defined as "chili maturity classification," which aims to simultaneously localize chili fruits and accurately classify them into specific maturity stages (FM, PM, and IM) under complex conditions. For consistency, this task is referred to as "maturity classification" throughout the remainder of this article. The overall technical roadmap is characterized by a high degree of systematicity and coherence, comprising four core phases: 1) dataset construction and preprocessing, 2) model architecture optimization, 3) visual interpretation and fine-grained classification, and 4) terminal deployment of the intelligent system.

To ensure data diversity and enhance model generalization capabilities, the research methodology is structured into four distinct phases. In the data acquisition and preprocessing phase, raw field images of Dongshan Guangpi chilies were collected under varying temporal conditions, illumination intensities, and shooting angles. Rigorous manual annotation was performed, followed by a hybrid augmentation strategy incorporating brightness adjustment, noise injection, and random rotation to construct a high-quality, task-specific dataset. For the model optimization phase, a lightweight detection model named CMD-YOLO was proposed using YOLOv11n as

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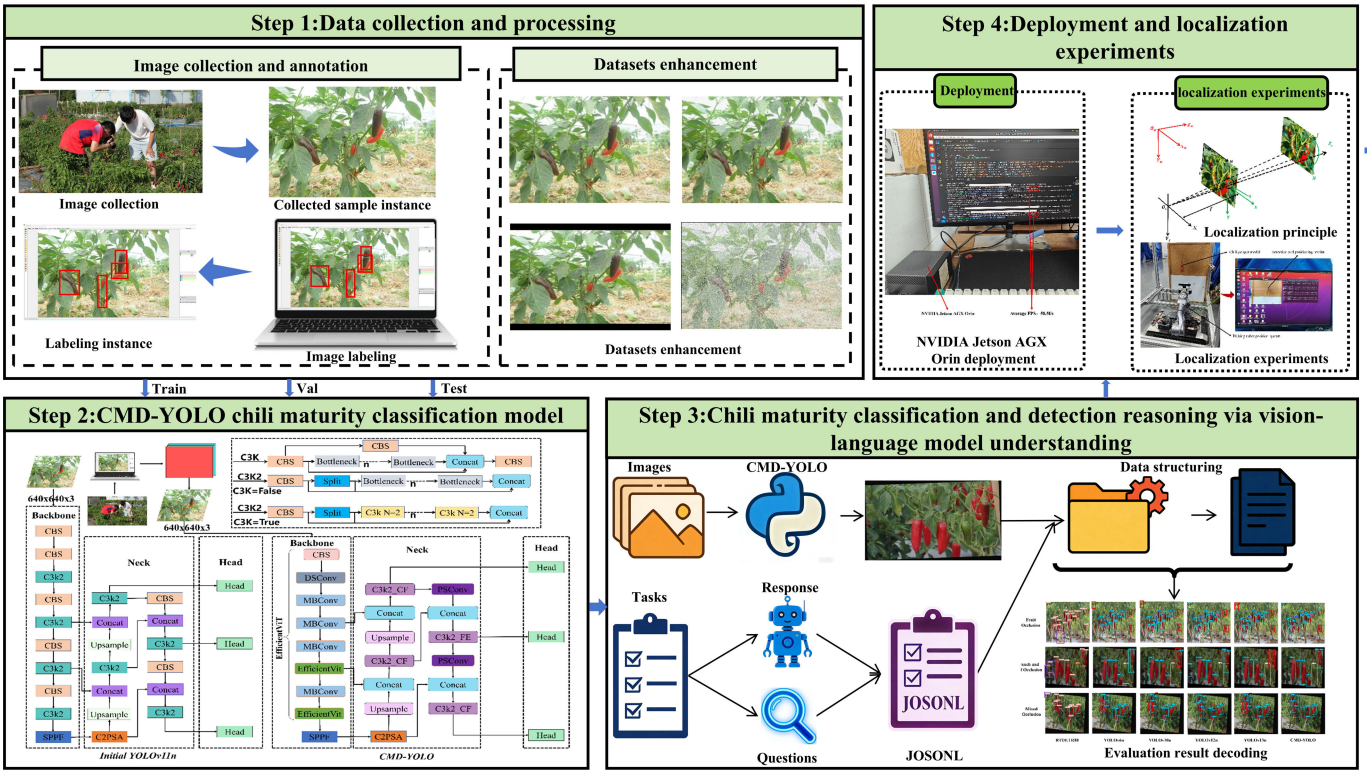


Fig. 3. Technical roadmap of chili fruit maturity classification.

the baseline. The architecture was systematically reconstructed by introducing the EfficientViT network as the backbone, integrating an improved C3k2_ConvFormer module into the neck network, and adopting a PSConv strategy for downsampling. In the training and evaluation phase, the model was trained to achieve precise classification across three maturity stages. Heatmaps were generated to verify that the network focused on discriminative fruit features rather than background noise. Finally, in the deployment and application phase, the optimal weights were deployed onto an NVIDIA Jetson AGX Orin embedded platform. Field harvesting trials integrating a picking robot were conducted to validate localization accuracy and harvesting success rates, providing reliable technical support for visual perception and decision-making in automated chili harvesting.

B. Chili Fruits Maturity Classification Model

1) Network Structure of YOLOv11: YOLOv11 adopts a three-stage architecture consisting of a backbone, neck, and detection head. Compared with earlier YOLO variants, particularly YOLOv8, YOLOv11 further optimizes the network architecture in terms of feature extraction, lightweight design, and multiscale feature fusion. Specifically, YOLOv11 introduces improved C3k2 modules to replace part of the conventional bottleneck structures, thereby enhancing local feature interaction while reducing computational redundancy. In addition, the newly incorporated position-sensitive attention (C2PSA) mechanism strengthens positional perception and contextual representation capability for critical target regions, improving feature discrimination under complex agricultural environments. Compared with previous YOLO versions, YOLOv11 also improves feature aggregation efficiency in the neck network and enhances the dynamic allocation capability of the detection head for targets at different scales, resulting in better robustness for small-object detection and dense target perception.

Benefiting from these architectural improvements, YOLOv11 achieves a more favorable balance between detection accuracy and computational complexity, making it more suitable for real-time deployment on edge devices in intelligent agricultural applications. In this study, the lightweight YOLOv11n variant is selected as the baseline model because it effectively reduces parameter quantity and computational overhead while maintaining competitive detection performance.

In the backbone network, YOLOv11 introduces optimized C3k2 modules and a position-sensitive attention (C2PSA) mechanism. The C3k2 module enhances local feature representation through feature partitioning and stacked 3×3 convolutions, while the C2PSA mechanism improves positional modeling capability for critical target regions. In the neck network, C3k2 modules are integrated into the PAN structure to strengthen multiscale feature aggregation and improve small-object perception capability. In addition, the detection head adopts a dynamic feature allocation strategy to improve detection performance for targets at different scales under complex scenes.

For chili maturity recognition tasks, YOLOv11n demonstrates favorable robustness under dense occlusion, illumination variation, and complex background interference.

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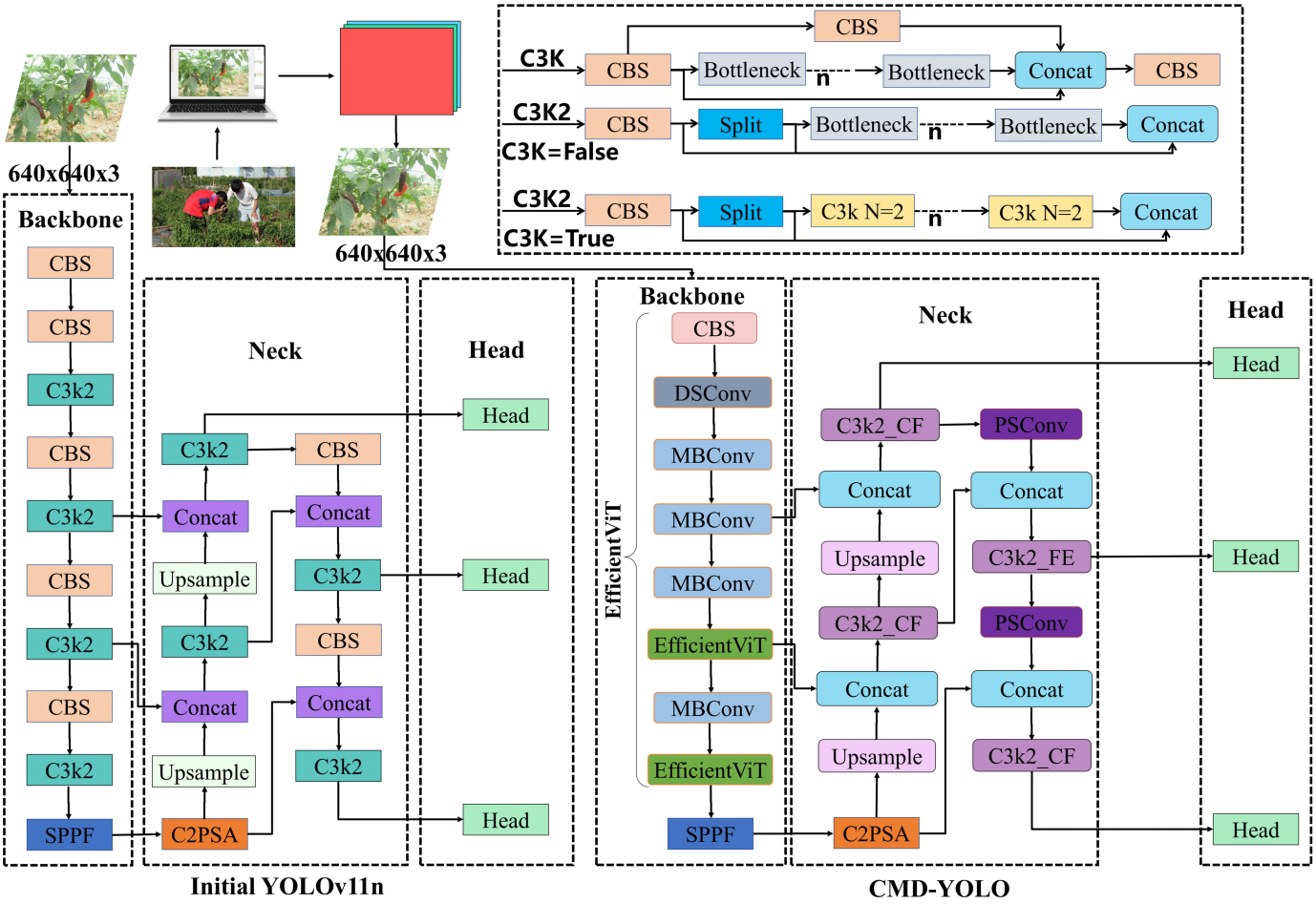


Fig. 4. Network structure of CMD-YOLO (improved YOLOv11n).

Through multiscale feature fusion and attention-guided feature representation learning, the model can effectively extract chromatic, morphological, and texture features of chili fruits at different maturity stages. Moreover, its lightweight architecture and high inference efficiency make it suitable for deployment on embedded agricultural robots and edge-computing devices.

As illustrated in Fig. 4, CMD-YOLO is reconstructed based on the YOLOv11n infrastructure through three systematic optimizations targeting the backbone, neck, and pooling layers.

1) *Lightweight Backbone Reconstruction:* To address the deployment difficulties on edge devices, the original CNN-based backbone of YOLOv11n is replaced by the EfficientViT network. Unlike the standard backbone, EfficientViT integrates attention mechanisms to augment global feature perception capabilities while significantly reducing parameter count and computational complexity.
2) *Enhanced Feature Fusion in Neck:* To overcome the low classification accuracy for small-sized chili fruits, a novel C3k2_ConvFormer module is designed and embedded within the neck network. This modification strengthens the fusion of features across different scales compared to the original C3k2 implementation, thereby enhancing the model's sensitivity to fine-grained morphological details.
3) *Multiscale Adaptability Optimization:* To maintain computational efficiency while preserving multiscale contextual information, the original spatial pyramid pooling fast (SPPF) module is retained in the backbone. Meanwhile, the proposed CMD-YOLO enhances multiscale feature representation mainly through the introduction of the C2PSA module and the redesigned feature fusion strategy in the neck. These improvements strengthen contextual information aggregation and cross-scale feature interaction, thereby improving the robustness of maturity classification for chili fruits with varying sizes, shapes, and complex occlusions.

2) EfficientViT Lightweight Network: To address the complex challenges of visual perception for chili maturity in unstructured field environments, while simultaneously satisfying the stringent hardware resource constraints of harvesting robots, this study reconstructs the feature extraction network using the EfficientViT hybrid architecture. As illustrated in Fig. 5, this design adheres to a hierarchical paradigm, effectively decoupling and reconstructing chili features from the macroarchitectural level down to micromechanisms.

In the overall architecture depicted in Fig. 5(a) and (b), the EfficientViT backbone adopts a hybrid strategy rather

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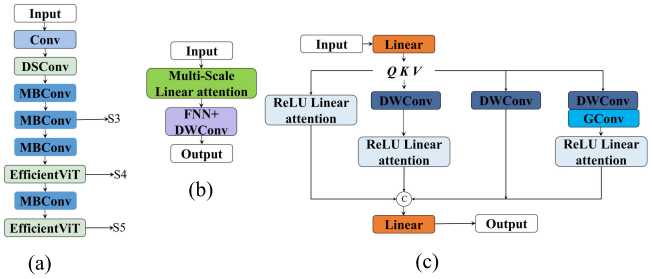


Fig. 5. EfficientViT backbone network structure diagram. (a) EfficientViT backbone. (b) EfficientViT block. (c) MSLA.

than the computationally expensive pure Vision Transformer (ViT) design. The shallow stages of the network utilize mobile inverted bottleneck convolution (MBConv) layers. This design choice is based on the theoretical premise that the early stages of visual processing rely heavily on local high-frequency information, including the sharp edges of chili fruits and stem textures. Convolution operations are inherently efficient at extracting these specific features. As the network depth increases, Transformer modules are introduced to capture long-range semantic dependencies. By modeling the global context, the model effectively distinguishes chili clusters from foliage backgrounds that exhibit similar coloration.

The core innovation within the network module lies in the multiscale linear attention (MSLA) mechanism shown in Fig. 5(c), which fundamentally resolves the computational bottleneck of standard Softmax attention that scales quadratically $O(N^2)$. This mechanism substitutes the Softmax operation with a hardware-friendly ReLU kernel function and reconstructs the computation path based on the associative property of matrix multiplication. From a mathematical perspective, given the projected query (Q), key (K), and value (V), the attention computation is reformulated as shown in the following equation:

$$\text{Attention}(Q, K, V) = \phi(Q)\left(\phi(K)^T V\right). \tag{1}$$

Crucially, to compensate for the inherent limitation of linear attention in processing local details, the module integrates a parallel depthwise convolution (DWConv) branch alongside the attention path. This dual-pathway design achieves multiscale feature aggregation: the linear attention branch focuses on capturing the global gradient patterns of fruit maturity (low-frequency information), while the DWConv branch is responsible for preserving fine-grained texture details (high-frequency information). The final feature output Y is a fusion of these complementary perspectives in the following equation:

$$Y = \text{Concat}(\text{LinearAttn}(Q, K, V), \text{DWConv}(V))\, W_o. \tag{2}$$

This design ensures that the model retains the robustness of global modeling without compromising the precision of local feature extraction, perfectly aligning with the high-standard requirements for intelligent harvesting in precision agriculture.

3) C3k2_ConvFormer Module: Accurately identifying subtle variations in color, texture, and morphology is crucial for chili maturity classification. However, traditional CNNs, such as the C3 module, excel at extracting local features but possess intrinsic limitations in capturing global image context. This deficiency may lead to inaccurate classification of chilies, particularly those in transitional maturity stages where visual cues are ambiguous. To resolve this issue, inspired by the cross-modal ConvFormer architecture, this article innovatively integrates the ConvFormer module into the standard C3k2 block to construct a novel C3k2_ConvFormer module. This design aims to retain the efficient local feature extraction capability of C3k2 while introducing a Transformer pathway based on self-calibrated convolutions and global attention mechanisms, thereby achieving effective modeling of multiscale, long-range dependencies in chili images. By learning local and global features in parallel, ConvFormer acquires more discriminative representations with lower resource consumption, enhances semantic discrimination, mitigates interclass confusion, and better captures the characteristics of chilies at different maturity levels. This improvement elevates maturity classification accuracy and provides the visual system of harvesting robots with highly robust core feature extraction and perception capabilities.

In the 3-D image scenarios applied to chili maturity classification, the CNN self-calibrated convolution module captures key features by leveraging weight sharing and information interaction among kernels. This structure effectively expands the receptive field through low-resolution calibration, avoiding the introduction of complex parameters and tedious hyperparameter tuning processes. The module utilizes lower-resolution feature maps (X_1) to generate attention weights that guide feature learning in the original image space, enabling each pixel to adaptively fuse long-range contextual semantic information, thus enhancing the extraction of discriminative features such as fruit color and texture. The network structure of the C3k2_ConvFormer module is illustrated in Fig. 6.

Specifically, the input feature X_i is first downsampled using a 3×3 2-D average-pooling with a stride of 4. Subsequently, it is processed by a 2-D convolution (3×3) and upsampled via bilinear interpolation. The result is added to the original feature X_i as a residual connection. where 0.5 c_{out} denotes half of the total number of output channels, that is, reducing the channel dimension by half. Finally, attention weights are obtained via Softmax normalization as shown in the following equation:

$$W = Softmax\left(Up\left(Conv_{3\times3}\left(Pool\left(X_1\right)\right)\right)\right) + X_1. \tag{3}$$

These weights are applied to each pixel position, and the enhanced feature X_2 is further generated through 2-D convolution. In the Transformer component, X_2 is mapped into query (Q), key (K), and value (V) vectors via linear layers, with average pooling used to compress the spatial dimensions of K and V to reduce computational complexity. A global attention map is computed using the scaled dot-product attention mechanism and applied to V, outputting a feature map X_3 with global contextual information as shown in the following equation:

$$X_3 = Softmax\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V \tag{4}$$

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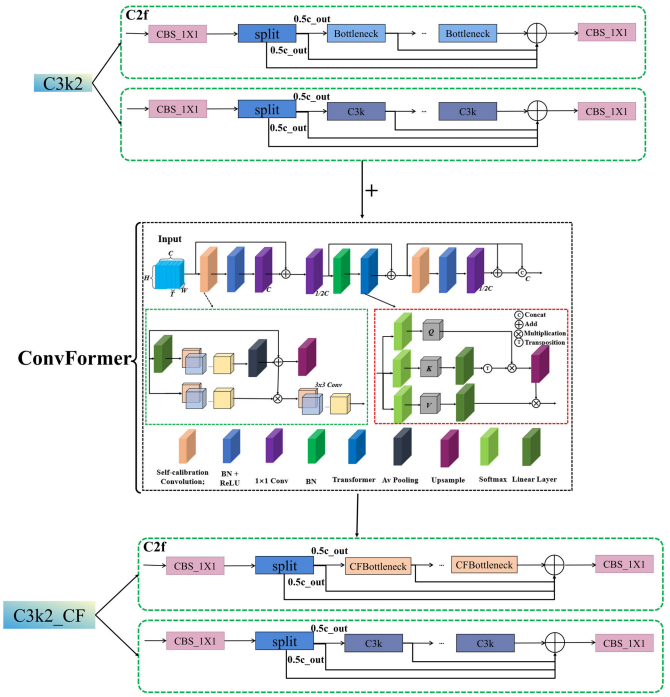


Fig. 6. Structure of C3k2_ConvFormer.

where d_k is the dimension of the key vector. The introduction of batch normalization (BN) and residual connections further improves the learning efficiency and generalization of global features. Finally, the output feature X_s is processed again by the CNN part to obtain X_t. By reusing and concatenating global features, the model enhances its expressive power while maintaining high computational efficiency. Consequently, the C3k2_ConvFormer extracts stronger discriminative representations with lower computational overhead by parallelizing the learning of local details and global context, thereby improving the model's ability to distinguish semantic information, effectively alleviating confusion between maturity categories, and more precisely capturing the subtle differences in chilies across various maturation stages.

4) PSConv Downsampling Convolution Module: While convolution operations serve as the cornerstone of feature extraction and downsampling, playing a pivotal role in reducing feature map dimensions, traditional convolutions exhibit significant limitations when applied to chili maturity classification in unstructured agricultural environments. The rapid reduction in spatial dimensions often precipitates a substantial loss of fine-grained textural details—such as the subtle color gradients during the transitional ripening phase—and edge information. This paucity of detail severely compromises the integrity of the model's receptive field coverage, thereby impeding accurate classification and localization of small, occluded chili fruits. To ameliorate this issue while simultaneously broadening the receptive field and capturing multiscale morphological features without compromising computational efficiency, this study introduces a plug-and-play PSConv module to reconstruct the downsampling process. The network structure of the PSConv module is illustrated in Fig. 7.

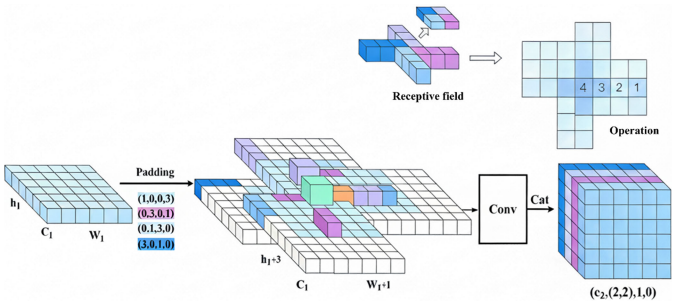


Fig. 7. PSConv downsampling convolution module structure diagram.

The integration of PSConv engenders a transformative enhancement in chili feature extraction, derived principally from its ingenious combination of nonuniform padding and multidirectional convolution kernels. Specifically, addressing the variable growth orientations and diverse morphologies of chili fruits, PSConv adaptively captures edge information across different orientations by applying distinct horizontal and vertical kernels [as formalized in (5)–(8)]

$$X_1^{(b',w',c')} = SiLU\left(BN\left(X_p^{(b,w,c_1)} \otimes W_1^{(1,3,c,c')}\right)\right) \tag{5}$$

$$X_2^{(b',w',c')} = SiLU\left(BN\left(X_p^{(b,w,c_1)} \otimes W_2^{(3,1,c,c')}\right)\right) \tag{6}$$

$$X_3^{(b',w',c')} = SiLU\left(BN\left(X_p^{(b,w,c_1)} \otimes W_3^{(1,3,c,c')}\right)\right) \tag{7}$$

$$X_4^{(b',w',c')} = SiLU\left(BN\left(X_p^{(b,w,c_1)} \otimes W_4^{(3,1,c,c')}\right)\right). \tag{8}$$

From the perspective of computational efficiency and model lightweighting, PSConv leverages a grouped convolution mechanism to significantly expand the effective receptive field while substantially reducing parameter count. As deduced from (9) and (10), PSConv achieves a measurable reduction in parameters compared to standard 3×3 convolutions. This characteristic is of paramount importance for deploying detection models on resource-constrained mobile harvesting robots or edge computing devices, enabling enhanced inference speeds without sacrificing detection accuracy

$$Y'\left(b_{b',w_2,c_2}\right) = SiLU\left(BN\left(X'\left(b',w',c'\right) \otimes W^{(2,2,c_1,c_2)}\right)\right) \tag{9}$$

$$Conv_{params} = c_2 \times c_1 \times k, \quad (bias = False). \tag{10}$$

Crucially, the convolutional response of PSConv exhibits characteristics akin to a Gaussian distribution, where convolution intensity diminishes from the center toward the periphery. This property is exceptionally valuable for chili maturity identification: it compels the model to focus more intensively on feature extraction from the fruit body (central region), accurately fitting the spatial distribution attributes of the fruit while effectively suppressing noise interference from irrelevant backgrounds such as surrounding foliage and soil. By integrating PSConv into the YOLOv11n model, the network not only strengthens its capability to extract underlying textural and color features but also improves localization precision for occluded or small-sized fruits, thereby significantly enhancing the robustness of maturity classification in complex field environments.

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TABLE I
TRAINING PARAMETERS

Training parameter	Value or type
Input image size	640×640×3
Optimizer	SGD
Optimizer momentum	0.937
Optimizer weight decay rate	0.0005
Initial learning rate	0.01
Final learning rate	0.01
Batch size	16
n	3
Epoch	300

TABLE II
COMPARATIVE TESTS WITH SOTA MODELS

Model	$P/\%$	$R/\%$	$mAP_{50}/\%$	$mAP_{50-95}/\%$	Parameters/M	FLOPs/G
Faster R-CNN [53]	73.0	86.3	87.2	58.6	44.26	85.0
SSD [54]	91.6	52.7	73.8	48.2	11.37	20.0
EfficientDet [55]	89.1	79.3	87.6	34.6	3.98	2.5
RTDETR-18 [65]	92.3	85.6	93.4	73.7	19.8	58.6
YOLOv5n [66]	89.3	86.2	93.1	72.9	2.50	8.4
YOLOv6n [67]	89.9	85.7	92.9	73.7	4.23	11.9
YOLOv8n [68]	90.1	86.3	93.0	74.0	3.01	8.2
YOLOv9t [69]	88.8	**88.3**	93.5	73.7	**1.97**	7.9
YOLOv10n [70]	91.4	85.1	93.7	74.1	2.70	8.4
YOLOv11n [71]	91.5	86.7	91.9	**76.8**	2.58	**6.4**
YOLOv12n [72]	89.4	88.2	94.0	74.6	2.56	6.5
YOLOv13n [73]	90.0	87.8	93.9	74.9	2.45	**6.4**
CMD-YOLO	**92.9**	86.1	**94.4**	75.3	3.66	8.1

IV. RESULTS AND DISCUSSION

To thoroughly validate the effectiveness of the CMD-YOLO model for chili fruit maturity classification, this study conducted a multidimensional experimental evaluation encompassing ablation studies [51], [52], comparative experiments [53], [54], visual detection analysis [55], [56], gradient-weighted class activation mapping (Grad-CAM) heatmap interpretation [57], [58], [59], and edge deployment testing [60], [61]. These assessments collectively examined the model's performance regarding detection accuracy, computational efficiency, and deployment feasibility [62], [63].

A. Test Environment and Evaluation Metrics

1) Test Environment and Parameters: To ensure the stability and reproducibility of the experiments, all model training and testing were conducted within a unified high-performance computing environment. The hardware configuration featured a 13th Gen Intel[1] Core[2] i7-13700KF processor (3.4 GHz) coupled with an NVIDIA GeForce RTX 3060 GPU, providing robust computational support for deep learning tasks. The operating system employed was Windows 10 Professional (64-bit) to ensure optimal hardware–software compatibility. In terms of the software ecosystem, algorithms were developed and implemented using the Python programming language based on the PyTorch deep learning framework, with PyCharm Community Edition serving as the integrated development environment. To eliminate potential interference from environmental variables, all ablation studies and comparative analyses were executed under this standardized protocol. The chili maturity dataset underwent uniform preprocessing before being fed into the various algorithms, thereby facilitating a systematic and quantitative assessment of model performance and optimization efficacy. The experimental training parameters are shown in Table I.

2) Model Evaluation Metrics and Training Protocols: To comprehensively evaluate model performance across dimensions of accuracy and efficiency, precision (P), recall (R), mean average precision (mAP), and parameter count were selected as the core evaluation metrics. The training process utilized the stochastic gradient descent (SGD) optimizer with an initial learning rate set to 0.01. A dynamic learning rate adjustment strategy based on cosine annealing was implemented, decaying at a rate of 0.9 every 10 epochs, to effectively balance convergence speed with generalization capability. The batch size was set to 16, and the training duration spanned 300 epochs. Crucially, to ensure scientific rigor and objective comparison, all hyperparameters were strictly standardized across the improved CMD-YOLO model and all contrast models. Furthermore, no pretrained weights were loaded; all models were trained from scratch to provide a reliable and unbiased basis for analyzing performance differences and validating algorithm optimization. The calculation formulas for each evaluation metric are given in (11)–(14):

$$P = \frac{C_{TP}}{C_{TP} + C_{FP}} \times 100\% \tag{11}$$

$$R = \frac{C_{TP}}{C_{TP} + C_{FN}} \times 100\% \tag{12}$$

$$AP = \int_0^1 P(R)\, dR \tag{13}$$

$$mAP = \frac{\sum_{i=1}^{n} AP_i}{n}. \tag{14}$$

In the proposed formulas, C_{TP} denotes the number of instances where the model correctly predicts the actual maturity stage of a given chili fruit. C_{FP} represents the count of instances where fruits from other maturity stages are incorrectly classified as belonging to the target stage. C_{FN} refers to the number of instances where the model fails to identify the correct maturity stage of a target fruit, misclassifying it into a different category. Furthermore, average precision (AP) quantifies the model's overall capability to accurately detect and discriminate among the various maturity stages of chili fruits.

B. Comparison Experiments Among Different Models

To comprehensively evaluate the overall performance of the improved CMD-YOLO model in chili fruit maturity classification, this study selected current mainstream object detection algorithms for comparative testing, including Faster R-CNN, EfficientDet, SSD, RT-DETR-18, YOLOv5n, YOLOv6n, YOLOv8n, YOLOv9t, YOLOv10n, YOLOv11n, YOLOv12n, YOLOv13n, and CMD-YOLO. All models were trained and tested under a unified dataset and hardware platform with consistent parameter configurations to ensure the objectivity and fairness of the results. As shown in Table II, the improved CMD-YOLO model demonstrated significant advantages in both lightweight design and detection accuracy. We

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did not add recently proposed agriculture-specific lightweight detectors such as Chili-YOLO and CES-YOLOv8 to the table for horizontal comparison. The primary reason is that these methods have not released complete official training codes and dataset configurations, and their experimental pipelines differ greatly from our unified training settings, which would introduce unfair evaluation if compared directly. To compensate for this limitation, we have supplemented comprehensive reviews of these agricultural detection algorithms in the related work section and added clear explanatory notes for the current experimental design.

Compared with the representative detection models listed in Table II, CMD-YOLO achieves a favorable balance between detection accuracy and lightweight deployment. Experimental results show that CMD-YOLO obtains the highest precision (92.9%) and mAP_{50} (94.4%) among all compared methods. Compared with RT-DETR-18, the proposed model achieves higher detection accuracy while significantly reducing model complexity, requiring only 3.66 M parameters and 8.1 G FLOPs, which demonstrates better suitability for edge-device deployment.

Traditional two-stage detectors such as Faster R-CNN exhibit substantially larger parameter scales (44.2 M) and computational overhead (85.0G FLOPs), while their detection performance remains inferior to CMD-YOLO. This indicates limited real-time capability and insufficient feature representation for dense and occluded agricultural targets. Although SSD maintains relatively low computational complexity, its recall drops to 52.7%, revealing weak perception capability for occluded and densely distributed objects. EfficientDet achieves relatively compact model complexity; however, its $mAP_{50–95}$ remains only 34.6%, indicating limited localization robustness under complex maturity recognition scenarios.

Among the YOLO-series detectors, YOLOv9t and YOLOv13n achieve smaller parameter sizes, but both exhibit lower detection accuracy than CMD-YOLO. Compared with YOLOv11n, CMD-YOLO improves precision and mAP_{50} by 1.4% and 2.5% points, respectively, demonstrating that the proposed feature fusion and lightweight optimization strategies effectively enhance small-object representation capability in complex agricultural environments. Although YOLOv11n achieves slightly higher $mAP_{50–95}$ performance, CMD-YOLO provides a more balanced tradeoff between overall detection accuracy and deployment efficiency.

As illustrated by the precision–recall (PR) curves in Fig. 8, CMD-YOLO consistently maintains high precision across different recall ranges, further confirming its detection robustness under challenging agricultural conditions. Overall, the proposed framework effectively controls computational complexity while maintaining competitive detection performance, making it more suitable for resource-constrained edge agricultural applications.

C. Visualization of Detection Results Across Different Models

The performance of object detection models is intrinsically linked to the quality and diversity of the training data. In facility agriculture environments, image acquisition for chili fruits is often comprised of complex factors such as

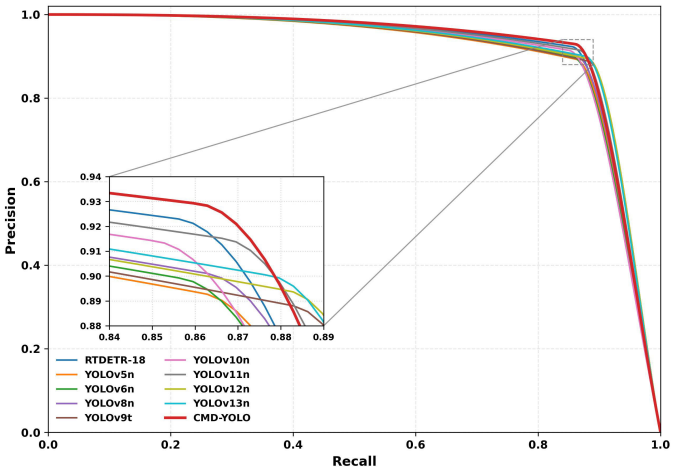


Fig. 8. PR curve comparison between the proposed CMD-YOLO algorithm and SOTA models. The proposed CMD-YOLO consistently maintains the highest precision across the entire recall range, demonstrating superior detection stability and robustness compared to other lightweight and advanced detection models.

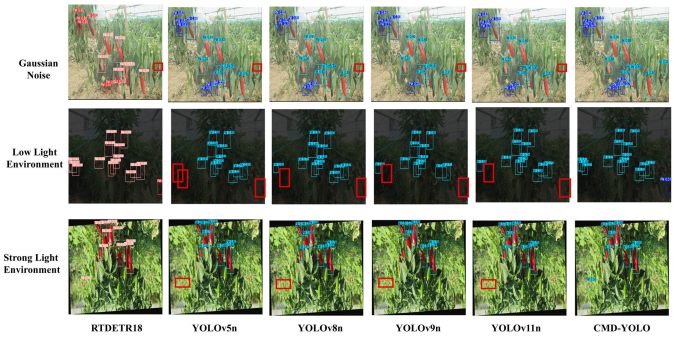


Fig. 9. Performance comparison of visual detection capabilities among contrast models under Gaussian noise and diverse illumination scenarios.

severe occlusion, uneven illumination, and background similarity (camouflaged targets). These factors frequently result in limited sample quantity and uneven distribution, thereby constraining the training efficacy of the model. To rigorously evaluate the robustness and generalization capabilities of the proposed method, comparative experiments were conducted using both the self-constructed Dongshan Guangpi chili test set and the public Kaggle chili maturity classification dataset. The visualization results under scenarios involving Gaussian noise, varying illumination intensities, and multilevel occlusion, as well as on the Kaggle dataset, are presented in Figs. 9–11. Note: the pink and blue boxes denote predicted cases, red boxes denote missed cases, and purple boxes denote misclassified cases.

From the detection results, the CMD-YOLO model demonstrated superior performance across diverse datasets. As illustrated in Fig. 9, under conditions of Gaussian noise and extreme lighting (strong/weak), comparison models such as RT-DETR-18, YOLOv5n, YOLOv8n, YOLOv9t, and YOLOv11n exhibited varying degrees of false negatives (missed detections), indicating a high sensitivity to illumination changes. In contrast, the improved CMD-YOLO model showed virtually no false detections, verifying its

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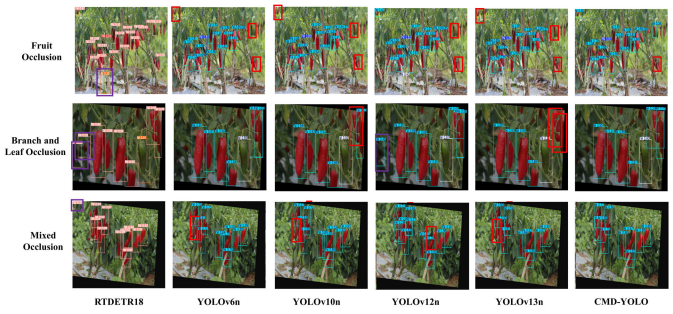


Fig. 10. Performance comparison of visualization detection capabilities across contrast models in various occlusion scenarios.

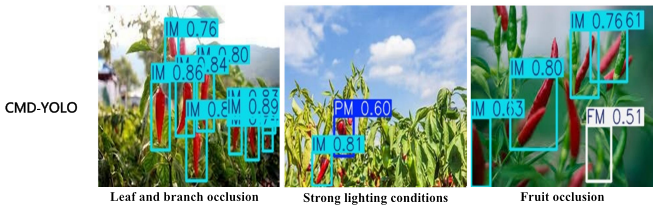


Fig. 11. Detection experiment on the publicly available chili fruit dataset from Kaggle.

robustness in maintaining stable maturity recognition under fluctuating light intensities. Furthermore, Fig. 10 depicts detection performance under complex occlusion scenarios. Models including RT-DETR-18, YOLOv6n, YOLOv10n, YOLOv12n, and YOLOv13n suffered from significant omission and commission errors. This failure is largely attributed to the severe occlusion of chili targets, which reduces them to small objects with sparse feature information, making feature extraction difficult for baseline models. Conversely, the proposed model only missed a few extremely blurred small targets with negligible false positives, demonstrating not only high robustness but also strong generalization in feature extraction under obstruction.

The generalization capability of the proposed model was further evaluated using the public Kaggle chili dataset, as illustrated in Fig. 11. Experimental results show that CMD-YOLO successfully detected most chili targets under unseen environmental conditions, demonstrating favorable adaptability to cross-domain data. On the self-constructed dataset, the model achieved high precision (P) and recall (R) with relatively few false detections and missed detections, indicating that the proposed framework effectively captures the visual characteristics of Dongshan Guangpi chilies in practical planting environments. When tested on the Kaggle dataset, the detection performance exhibited a slight decrease compared with the self-constructed dataset due to differences in chili varieties, imaging conditions, and scene distributions. Nevertheless, without additional retraining or domain-specific fine-tuning, CMD-YOLO still maintained competitive detection performance, reflecting its inherent feature generalization capability. Overall, the proposed framework demonstrates good adaptability and robustness under different agricultural scenarios, providing reliable visual perception support for intelligent harvesting applications.

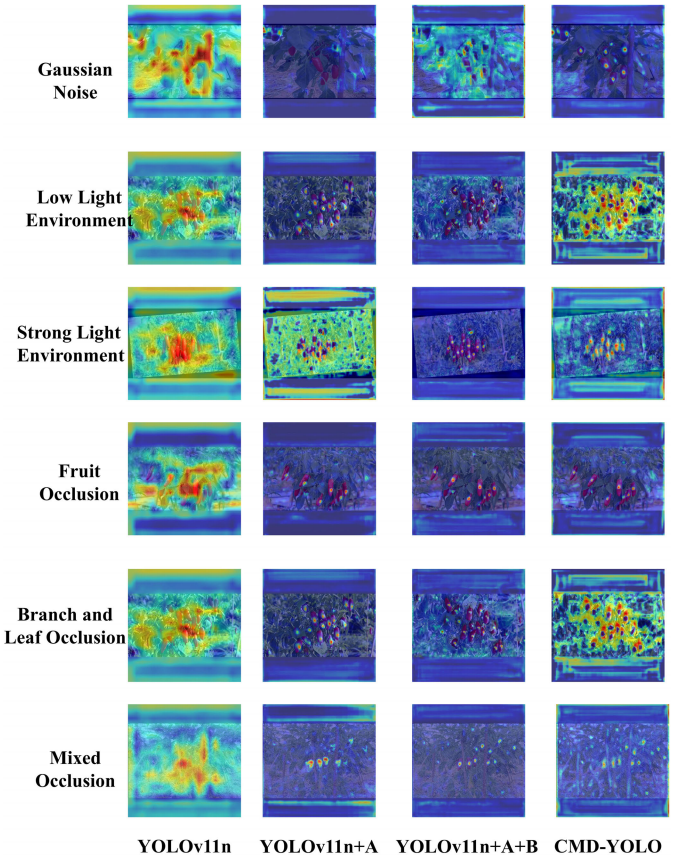


Fig. 12. Comparison of heatmap detection effect of each model in different scenarios.

TABLE III
DEPLOYMENT COMPARISON OF LIGHTWEIGHT MODELS ON THE NVIDIA JETSON AGX ORIN PLATFORM

Model	Parameters/M	FLOPs/G	FPS/(f/s)	Latency/ms
EfficientDet	3.98	2.5	48.2	20.7
YOLOv8n	3.01	8.1	66.2	15.1
YOLOv10n	2.27	6.5-	54.2	18.5
YOLOv11n	2.58	6.3	51.9	19.3
YOLOv12n	2.56	6.3	35.9	27.9
CMD-YOLO	3.66	8.1	58.5	17.1

D. Heatmap Visualization Analysis

To provide interpretability for the improved YOLOv11n-based maturity classification, Grad-CAM heatmaps were utilized to visualize the model's focus on fruit features. Heatmaps intuitively reflect the network's ability to extract and attend to maturity-related characteristics. A comparative analysis of heatmaps between the CMD-YOLO model and the original YOLOv11n was conducted across different scenarios, as shown in Fig. 12. For Dongshan Guangpi chilies at various maturity stages (green, color-turning, and red), the heatmaps generated by the original model exhibited dispersed and unfocused high-activation regions, indicating a susceptibility to interference from environmental noise, lighting, and background clutter.

From the perspective of detection performance, the heatmap visualization further verifies the effectiveness of the proposed

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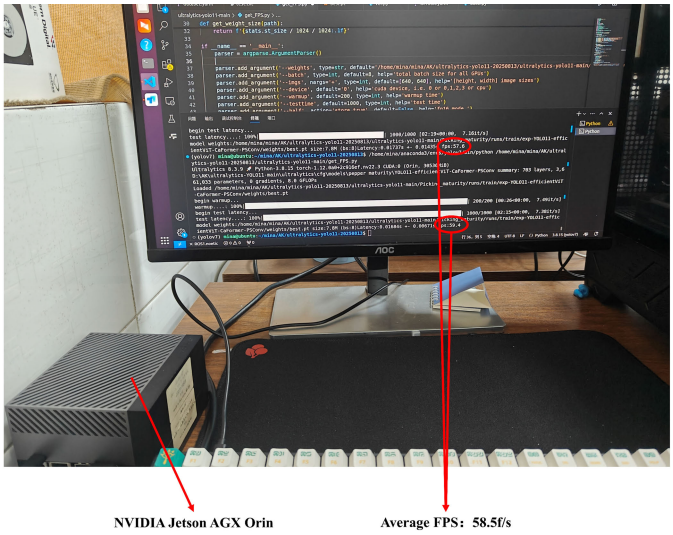


Fig. 13. NVIDIA Jetson AGX Orin deployment experiments.

framework. The activated regions are mainly concentrated on the color and texture characteristics closely related to chili maturity, indicating that the model can effectively focus on discriminative target features. This demonstrates that the integration of the EfficientViT backbone, the C3k2_ConvFormer module, and the PSConv strategy improves the representation capability of maturity-related features. In complex agricultural environments, the heatmaps remain stably focused on the target regions even under partial occlusion and background interference. This suggests that the optimized feature extraction and downsampling structures enhance the robustness of the model against environmental noise and illumination variation, thereby improving detection stability. In addition, while maintaining accurate feature perception capability, the proposed framework preserves relatively low computational complexity and parameter scale, which is beneficial for real-time deployment on edge devices. Nevertheless, in scenarios involving severe occlusion, where the target fruit is heavily concealed by branches or leaves, the activated regions become relatively dispersed. This indicates that the current framework still has limitations in extracting highly discriminative features under extreme occlusion conditions. Future work will further investigate more effective context modeling and feature aggregation strategies to improve perception robustness in highly complex agricultural environments.

E. Deployment Experiments

To further evaluate the deployment performance of the proposed model on edge devices and its applicability to chili harvesting scenarios, the trained CMD-YOLO model was deployed on the NVIDIA Jetson AGX Orin platform, as illustrated in Fig. 13, and real-time inference experiments were conducted. In addition, based on the lightweight model comparison results presented in Table III, and considering that models such as SSD and Faster R-CNN involve relatively large parameter scales that are unsuitable for real-time edge deployment, additional deployment experiments were performed using several lightweight models, including EfficientDet, YOLOv8, YOLOv10, YOLOv11, and YOLOv12, under the same hardware environment. Comprehensive comparisons were conducted in terms of parameter size, FLOPs, and inference speed.

The edge deployment results reveal substantial differences among lightweight models in inference efficiency and computational complexity. Although EfficientDet maintains relatively low computational cost, its inference speed reaches only 48.2 f/s, indicating that the parallel execution efficiency of its feature fusion architecture remains limited on the Jetson AGX Orin platform. YOLOv8n achieves the highest inference speed of 66.2 f/s together with the lowest latency of 15.1 ms, demonstrating strong real-time capability. However, combined with the previous detection results, its overall detection accuracy remains lower than that of CMD-YOLO. YOLOv10n and YOLOv11n preserve favorable lightweight characteristics in parameter scale and inference efficiency. In particular, YOLOv11n achieves an inference speed of 51.9 f/s with only 2.58 M parameters and 6.3 G FLOPs, demonstrating high deployment efficiency. Nevertheless, its detection accuracy and small-object representation capability under complex field environments remain inferior to those of CMD-YOLO. In comparison, although YOLOv12n maintains a parameter scale and computational cost similar to YOLOv11n, its inference speed decreases to 35.9 f/s, and the latency increases to 27.9 ms, suggesting that its network architecture introduces a heavier inference burden on edge devices and consequently reduces practical deployment efficiency.

Overall, CMD-YOLO achieves a real-time inference speed of 58.5 f/s with a latency of 17.1 ms while maintaining superior detection accuracy, demonstrating more balanced overall performance than most comparative models. The deployed CMD-YOLO model contains 3.66 M parameters with a computational cost of 8.1 G FLOPs. Although its parameter scale and computational complexity are slightly higher than those of several highly compressed models, the additional computational overhead provides more stable small-object detection capability and stronger detection performance under complex field environments. Compared with YOLOv11n, CMD-YOLO achieves higher detection accuracy and more robust performance in complex scenes with only an additional increase of approximately 1.08 M parameters. These results indicate that the proposed framework achieves a favorable balance among detection accuracy, real-time inference capability, and deployment efficiency, making it more suitable for agricultural robotic vision systems operating under resource-constrained edge environments.

F. Positioning Method and Experiments

1) Visual Geometry-Based Positioning Theoretical Analysis: To guide the robotic manipulator's end-effector precisely to the chili-harvesting target, a mapping model from the 2-D image plane to the 3-D operational space must be established. As theoretically illustrated in the coordinate transformation logic (referencing the geometric principles in Fig. 14), the chili centroid J and the edge reference point N are first extracted in the pixel coordinate system O_{nv} via the target detection

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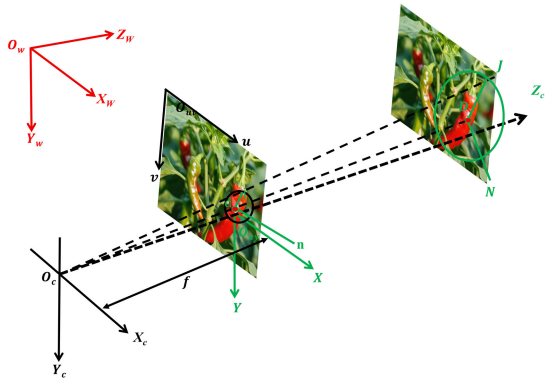


Fig. 14. Core coordinate system transformation relationship in the chili pepper visual localization process, including the world coordinate system (O_W-X_W-Y_W-Z_W), camera coordinate system (O_C-X_C-Y_C-Z_C), and image coordinate system (O-u-v).

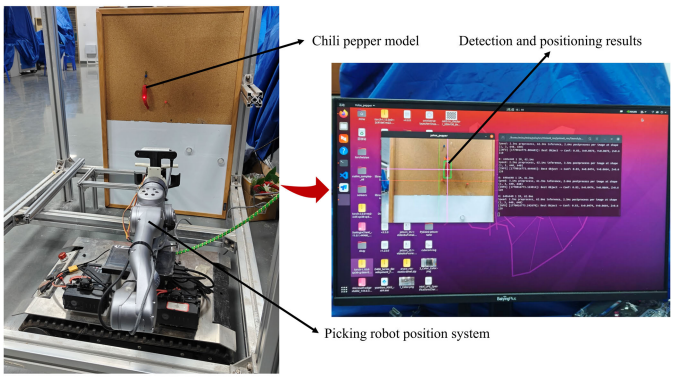


Fig. 15. Indoor simulated experimental scene of the localization system for chili pepper picking robots. The experimental platform mainly consists of a chili pepper model (simulating real chili targets in the field) and a picking robot localization system (integrating visual perception and mechanical execution units). The terminal on the right displays the detection and positioning results of chili pepper targets in real time, accurately identifies the target position with reddish-green localization boxes, and outputs core positioning parameters. This platform is used to verify the accuracy of the chili pepper localization algorithm and the response performance of the robot system, providing an indoor test basis for subsequent actual field operations.

network. To convert the pixel distance in the image into the actual operational distance in physical space, a transformation from the pixel coordinate system to the world coordinate system O_w is essential.

Let J denote the true centroid of the chili pepper and N be the edge reference point in the world coordinate system. The physical distance between them is defined as D (representing the target geometric dimension). Let f be the focal length, representing the distance from the depth camera's optical center O_c to the imaging plane, and let Z be the vertical depth distance from O_c to the target fruit plane. Based on the pinhole imaging principle and the similarity of triangles, the scaling factor P between the physical scale in the world frame and the scale on the imaging plane can be derived as

$$P = \frac{Z}{f} \tag{15}$$

where Z is obtained directly from the depth camera (mm) and f is the intrinsic focal length (mm). In the image physical coordinate system O_{xy}, the projected length $d_{O_{xy}}$ between the pollination point and the reference point is expressed as

$$d_{O_{xy}} = \frac{D}{P} = \frac{D \cdot f}{Z}. \tag{16}$$

Furthermore, assuming the physical size of a single pixel on the image sensor is μ (mm/pixel), the corresponding pixel distance d (pixels) in the pixel coordinate system O_{uv} is calculated as

$$d = \frac{d_{O_{xy}}}{\mu} = \frac{D \cdot f}{Z \cdot \mu}. \tag{17}$$

Through this geometric model, the system can dynamically calculate the target pixel coordinates based on the preset picking radius D and the real-time depth information Z. This allows for the resolution of the 3-D picking vector in the world coordinate system, providing precise end-effector pose inputs for the manipulator's inverse kinematics solver.

2) Positioning Accuracy Experiment: To rigorously validate the synergistic precision between the visual perception system and robotic motion control, positioning accuracy trials were conducted at Laboratory 114, South of the 8th Teaching Building, Hunan Agricultural University (as shown in Fig. 15).

The experimental setup integrates depth information from the Intel RealSense D435 camera with the spatial constraints defined by the hand–eye calibration matrix. The target chili peppers were arranged in different spatial sectors (including near center, top-left, top-right, bottom-left, and bottom-right) within the manipulator's workspace to evaluate the global positioning consistency. Data from 14 independent grasping trials were collected, comparing the algorithm-calculated coordinates ($x_{algo}, y_{algo}, z_{algo}$) against the actual physical measurements ($x_{real}, y_{real}, z_{real}$). The positioning experimental data are shown in Table IV.

The quantitative statistical analysis reveals a distinct anisotropic characteristic in the system's positioning error. As detailed in the data collection table, the system demonstrated exceptional precision in the depth dimension, with a mean absolute error (MAE) of 0.0044 m (4.4 mm) along the Z-axis. This indicates that the depth estimation logic based on the bounding box center is highly reliable for determining the harvesting distance. However, the lateral positioning exhibited higher variability, with mean errors of 0.0808 m for the X-axis and 0.0858 m for the Y-axis. It is noteworthy that these aggregated lateral errors were significantly skewed by specific outliers in the peripheral regions (e.g., the top-left sector), whereas the positioning accuracy in the central workspace remained within the millimeter-level tolerance required for successful grasping.

Further analysis of the error distribution across different spatial quadrants reveals that the positioning deviation increases significantly as the target moves from the image center to the peripheral regions (top-left, bottom-right, etc.). This phenomenon of spatial nonuniformity in error can be attributed to three primary factors: the inherent optical characteristics of the imaging system, the propagation of cumulative residuals in the hand–eye calibration matrix, and the interdependence between depth measurement stability and lateral coordinate estimation.

The primary source of the observed spatial nonuniformity in positioning error arises from the inherent optical

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TABLE IV
LOCALIZATION EXPERIMENT USING THE CMD-YOLO DETECTION MODEL

ID	Fruit position	D435 camera-based detection and localization					
		x_{algo}/m	y_{algo}/m	z_{algo}/m	x_{real}/m	y_{real}/m	z_{real}/m
1	Near the center	0.0202	0.0198	0.5830	0.0240	0.0130	0.5870
2		0.0130	0.0623	0.6020	0.0150	0.0590	0.6060
3		0.0120	0.1244	0.6040	0.0140	0.1250	0.6090
4		0.0828	0.0223	0.5970	0.0900	0.0240	0.6000
5		0.1347	0.0889	0.6140	-0.1430	-0.0930	0.6180
6	Near the top-left	0.1704	0.1592	0.6110	0.1740	0.1580	0.6170
7		0.1076	0.0855	0.6040	0.1040	0.0820	0.6020
8	Near the top-right	0.0862	0.0023	0.5810	0.0860	0.0090	0.5870
9		0.1466	0.0361	0.5810	0.1500	0.0310	0.5850
10		0.1695	0.1043	0.5810	0.1750	0.1000	0.5870
11	Near the bottom-left	0.1316	0.1754	0.5820	0.1390	0.1840	0.5880
12		0.0708	0.1525	0.5720	0.0770	0.1600	0.5790
13	Near the bottom-right	0.0990	0.1611	0.5560	0.1010	0.1690	0.5620
14		0.1065	0.1465	0.5820	0.1090	0.1560	0.5910
Mean Error	X-axis	0.0808					
	Y-axis	0.0858					
	Z-axis	0.0044					

characteristics of the imaging system. While the coordinate resolution relies on a standard pinhole camera model, the RGB-D sensor inevitably exhibits radial lens distortion, particularly the "barrel effect" manifested at the periphery of the field of view. This optical nonlinearity causes the raw pixel coordinates to deviate from their ideal geometric projections. Consequently, when targets such as those in the "top-left" or "bottom-right" sectors are mapped to the 3-D space, these minute pixel-level deviations are linearly amplified by the depth factor, resulting in a pronounced degradation of lateral accuracy compared to the central workspace.

Furthermore, the propagation of cumulative residuals within the hand–eye calibration matrix contributes significantly to the error amplification in distal regions. The transformation matrix, obtained through global optimization algorithms, represents a best-fit solution that minimizes the average reprojection error across the calibration volume. However, linear transformation errors possess a tendency to propagate and amplify at the kinematic boundaries of the coordinate system. As a result, targets located in peripheral zones, which correspond to the distal ends of the manipulator's reach, are more susceptible to these cumulative geometric residuals, leading to the observed stochastic fluctuations in the *X*- and *Y*-axes for outliers in the dataset.

Finally, the interdependence between depth measurement stability and lateral coordinate estimation introduces a compounding error factor at the image boundaries. The stereo matching algorithm of the depth camera typically exhibits reduced signal-to-noise ratio at the edges of the disparity map due to geometric dilution of precision. Since the calculation of lateral coordinates is mathematically coupled with the depth value, any fluctuation or noise in the depth measurement at the image periphery directly propagates to the lateral coordinate estimation. This coupling effect explains why the *Z*-axis error remains relatively low globally, while the lateral errors exhibit significant spikes in regions where the depth map quality marginally deteriorates.

G. Ablation Experiment

Specifically, to verify the individual contributions and synergistic effects of the three proposed improvement modules, a systematic ablation study was performed using YOLOv11n as the baseline. The impacts of the EfficientViT backbone (Module A), the C3k2_ConvFormer neck module (Module B), and the PSConv downsampling module (Module C) were rigorously evaluated. The comparative analysis focused on key performance indicators, including detection accuracy (*P*, *R*, mAP_{50}, and $mAP_{50–95}$), and model complexity (parameters). The quantitative results are presented in Table V.

The ablation results in Table V demonstrate the effectiveness of the proposed modules in improving the detection performance of CMD-YOLO, while the newly supplemented FLOPs/G, FPS (f/s), and latency/ms further quantify the computational overhead and real-time inference delay brought by each module for edge IoT deployment. After introducing Module A, the model recall increased from 86.7% to 87.4%, while mAP_{50} improved by 2.4% points. Meanwhile, FLOPs rose moderately from 6.4G to 7.9G, FPS dropped from 323.0 to 99.5 f/s, and the corresponding inference latency increased sharply from 3.1 to 10.1 ms. These results indicate that Module A enhances the representation ability for small and partially occluded chili targets under complex field conditions, yet it brings a noticeable computational burden and an obvious rise in inference delay. Although precision decreased slightly, the overall detection capability was improved.

After further integrating Module B, recall increased to 88.6% and mAP_{50} reached the peak value of 94.5%. The FLOPs only slightly climbed to 8.1G, FPS marginally declined to 94.7 f/s with a tiny increment of model parameters, and latency merely rose by 0.5–10.6 ms. This suggests that Module B effectively strengthens multiscale feature interaction and improves feature aggregation for dense targets without introducing high extra computational cost and inference delay, achieving a favorable balance between detection accuracy and inference efficiency.

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TABLE V
CMD-YOLO ABLATION EXPERIMENT RESULTS

Models	Improved module			Metrics							
	A	B	C	P/%	R/%	mAP$_{50}$/%	mAP$_{50-95}$/%	FLOPs/G	FPS(f/s)	Latency/ms	Parameters/M
✓	–	–	–	91.5	86.7	91.9	76.8	6.4	**323.0**	**3.1**	**2.58**
✓	✓	–	–	91.0	87.4	94.3	74.6	7.9	99.5	10.1	3.74
✓	✓	✓	–	90.2	**88.6**	**94.5**	75.3	8.1	94.7	10.6	3.81
✓	✓	✓	✓	**92.9**	86.1	94.4	**75.3**	8.1	92.0	10.9	3.66

TABLE VI
PERFORMANCE COMPARISON OF TEST RESULTS FOR REPLACING DIFFERENT BACKBONE NETWORK MODULES

Replaced module	P/%	R/%	mAP$_{50}$/%	mAP$_{50-95}$/%	Parameters/M
Initial backbone	91.5	86.7	91.9	**76.8**	2.58
StarNet	**92.7**	83.9	93.7	72.8	**1.94**
MobileNetv4	90.7	85.0	93.5	73.0	5.43
EfficientViT	91.0	**87.4**	**94.3**	74.6	3.74

When Module C was incorporated together with Modules A and B, the final model achieved 92.9% precision and 94.4% mAP$_{50}$, retaining the same 8.1G FLOPs, 92.0 f/s, and 10.9-ms latency with only 3.66 M parameters. Compared with the baseline model without any improved modules, precision and mAP$_{50}$ improved by 1.4% and 2.5% points, respectively, while the growth of computational complexity and inference latency remained controlled within a small range. These results demonstrate that the three proposed modules complement each other effectively. Module C optimizes feature compression without extra computational overhead and delay growth, enabling the model to maintain high detection accuracy while preserving lightweight and real-time deployment capability on edge hardware.

A slight fluctuation in recall was observed for PM chili fruits. This is mainly because fruits at the transitional maturity stage exhibit ambiguous color and texture characteristics under illumination variation and shadow occlusion. Their visual features are easily confused with branches and foliage in complex agricultural environments. Nevertheless, the integration of the three feature optimization modules significantly alleviated this issue by improving semantic representation and contextual perception capability, and the model still maintains acceptable inference latency and running speed for edge robotic harvesting applications.

1) Effect of the EfficientViT Backbone: To systematically evaluate the impact of integrating specific modules at different network stages on chili fruit maturity classification performance, this study designed four sets of comparative experiments based on the YOLOv11n architecture. These experiments involved replacing the original backbone, modifying the C3k2 module, and substituting various lightweight up/downsampling convolution modules, with performance benchmarked against the original model.

The quantitative results in Table VI show that different backbone networks exhibit distinct tradeoffs between detection accuracy and model complexity. StarNet achieved the smallest parameter count of 1.94 M and relatively high precision, but its recall remained limited at 83.9%, indicating insufficient perception capability for densely distributed or partially occluded targets. In agricultural harvesting scenarios, missed detections may directly affect harvesting efficiency and yield.

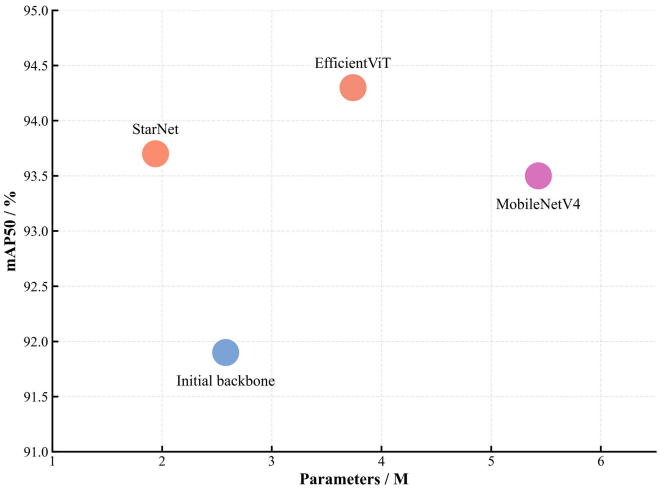


Fig. 16. Comparison of mAP$_{50}$ and parameter count among different backbone network replacements. The EfficientViT backbone achieves the highest mAP$_{50}$ of 94.4% with a moderate parameter scale, demonstrating the optimal tradeoff between detection accuracy and model lightweightness.

MobileNetV4 introduced the largest parameter scale among the compared lightweight backbones, while its detection performance showed limited improvement. This suggests that excessive lightweight compression may weaken the extraction of fine-grained spatial features required for small-object recognition.

In comparison, EfficientViT achieved the best overall balance among recall, localization accuracy, and model complexity. The model reached 94.4% mAP$_{50}$ and 74.6% mAP$_{50-95}$ while maintaining a lightweight structure. By combining convolutional local perception with global attention mechanisms, EfficientViT effectively improved feature robustness under occlusion, illumination variation, and dense target distribution, making it more suitable for edge-oriented agricultural perception tasks.

Fig. 16 further illustrates the relationship between model accuracy and parameter count. EfficientViT occupies a favorable position in the performance–complexity space, achieving higher detection accuracy with moderate computational cost. Therefore, EfficientViT was selected as the backbone network of CMD-YOLO.

2) Effect of the C3k2_ConvFormer Module: To comprehensively assess the effects of fusing different modules with

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TABLE VII
COMPARISON OF THE PERFORMANCE OF THE EXPERIMENTAL RESULTS OF REPLACING DIFFERENT MODULES FUSED WITH C3K2

Replaced module	$P/\%$	$R/\%$	$\text{mAP}_{50}/\%$	$\text{mAP}_{50-95}/\%$	Parameters/M
YOLOv11n+A	91.0	87.4	94.3	74.6	3.74
C3k2_EMA	88.4	**88.7**	93.9	73.9	3.78
C3k2_Faster	**93.3**	84.6	93.6	73.9	**3.59**
C3k2_Faster_EMA	90.2	88.5	94.0	74.2	**3.59**
C3k2_IdentityFormer	92.5	84.9	94.1	74.3	3.68
C3k2_PoolingFormer	91.6	86.0	93.6	74.6	3.68
C3k2_ConvFormer	90.2	88.6	**94.5**	**75.3**	3.81

TABLE VIII
PERFORMANCE COMPARISON OF EXPERIMENTS WITH DIFFERENT UPPER/LOWER SAMPLING CONVOLUTION MODULES

Replaced module	$P/\%$	$R/\%$	$\text{mAP}_{50}/\%$	$\text{mAP}_{50-95}/\%$	Parameters/M
YOLOv11n+A+B	90.2	**88.6**	**94.5**	75.3	3.81
Adown	92.0	86.3	94.2	74.8	3.68
ContextGuidedDown	90.8	86.1	93.5	74.7	4.22
CARAFE	90.5	86.8	93.6	74.6	3.95
Dysample	90.6	87.5	94.3	**75.6**	3.82
SPDConv	89.1	87.9	94.0	74.7	4.36
V7Downsample	91.3	86.1	94.2	75.1	3.69
PSConv	**92.9**	86.1	94.4	75.3	**3.66**

the C3k2 structure on detection performance, multiple comparative experiments were conducted based on the *YOLOv11n* framework.

The comparison results of different C3k2-based modules are presented in Table VII. The C3k2_EMA module improved recall and enhanced feature robustness, but precision decreased slightly. C3k2_Faster significantly improved precision while reducing parameter count, demonstrating favorable lightweight characteristics.

Among all compared variants, C3k2_ConvFormer achieved the best overall performance, reaching 94.5% mAP_{50} and 75.3% mAP_{50-95}. The results indicate that integrating convolutional operations with Transformer-based feature interaction effectively improves contextual modeling capability and enhances feature representation for small and occluded chili fruits.

In addition, C3k2_IdentityFormer and C3k2_PoolingFormer also achieved relatively balanced performance, indicating that hybrid feature interaction strategies contribute positively to feature fusion and detection stability. Overall, the proposed C3k2_ConvFormer module provides the most effective balance between detection accuracy and computational efficiency.

3) Effect of the PSConv Downsampling Convolution Module: To systematically evaluate the influence of different upsampling and downsampling convolution modules on chili fruit maturity classification performance, a series of comparative experiments were designed based on the *YOLOv11n* architecture. Various sampling modules were substituted into the network, and their performance was benchmarked against the original model.

The experimental results in Table VIII indicate that different downsampling strategies have a considerable influence on detection accuracy and lightweight performance. Adown achieved high precision with relatively low parameter complexity, but recall decreased noticeably, indicating reduced sensitivity to difficult targets.

ContextGuidedDown increased model complexity without achieving corresponding performance gains, while CARAFE maintained relatively balanced detection results but showed limited improvement over the baseline model.

Dysample achieved the highest recall and favorable mAP_{50-95} performance, demonstrating strong target perception capability. However, its parameter count increased slightly. SPDConv improved recall but introduced higher computational complexity and lower precision, limiting its deployment efficiency on edge devices.

Among all compared methods, PSConv achieved the best overall balance between detection accuracy and lightweight design. The module reached the highest precision of 92.9% while maintaining the lowest parameter count of 3.66 M. By preserving fine-grained spatial information during feature downsampling, PSConv effectively improved small-target perception and localization stability under complex field conditions.

Based on the above analyses, the final CMD-YOLO framework integrates EfficientViT, C3k2_ConvFormer, and PSConv to jointly enhance feature extraction capability, contextual representation, and lightweight deployment efficiency for chili maturity detection in edge-enabled agricultural systems.

V. CONCLUSION AND FUTURE WORK

This study proposed CMD-YOLO, a lightweight detection framework for multistage chili maturity recognition and robotic localization in complex agricultural environments. By integrating the EfficientViT backbone, the C3k2_ConvFormer module, and the PSConv downsampling strategy, the proposed method improved feature representation for small and occluded targets while preserving fine-grained spatial information. Experimental results demonstrated that CMD-YOLO achieved 94.4% mAP_{50} with only 3.66 M parameters and maintained real-time inference at 58.5 f/s on the NVIDIA Jetson AGX Orin platform. In robotic localization experiments, the proposed framework achieved a depth estimation error of 4.4 mm, indicating favorable spatial perception capability for robotic harvesting tasks.

The results demonstrate that CMD-YOLO provides an effective balance between detection accuracy, lightweight design, and edge deployment efficiency for IoT-enabled agricultural systems. Future work will focus on improving localization robustness through calibration optimization and multimodal sensor fusion, as well as integrating visual perception with autonomous robotic control strategies for fully automated

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harvesting applications. Once the complete official training codes and unified experimental configurations of agriculture-oriented lightweight detectors such as Chili-YOLO and CES-YOLOv8 are publicly available, we will unify experimental conditions and conduct comprehensive horizontal comparison experiments to further verify the superiority of CMD-YOLO in field chili detection tasks. In addition, future studies will further investigate multiplatform edge deployment, including energy consumption analysis, memory usage evaluation, GPU resource utilization, and system-level real-time performance on low-cost embedded devices. These investigations will help comprehensively validate the engineering applicability and deployment adaptability of CMD-YOLO in practical agricultural environments.

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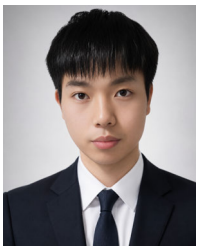
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